

7. The techniques of voltage/current division, series/parallel combination of impedance/admittance, circuit reduction, and Y - Δ transformation all apply to ac circuit analysis.
8. AC circuits are applied in phase-shifters and bridges.

Review Questions

- 9.1** Which of the following is *not* a right way to express the sinusoid $A \cos \omega t$?
- (a) $A \cos 2\pi ft$ (b) $A \cos(2\pi t/T)$
 (c) $A \cos \omega(t - T)$ (d) $A \sin(\omega t - 90^\circ)$
- 9.2** A function that repeats itself after fixed intervals is said to be:
- (a) a phasor (b) harmonic
 (c) periodic (d) reactive
- 9.3** Which of these frequencies has the shorter period?
- (a) 1 krad/s (b) 1 kHz
- 9.4** If $v_1 = 30 \sin(\omega t + 10^\circ)$ and $v_2 = 20 \sin(\omega t + 50^\circ)$, which of these statements are true?
- (a) v_1 leads v_2 (b) v_2 leads v_1
 (c) v_2 lags v_1 (d) v_1 lags v_2
 (e) v_1 and v_2 are in phase
- 9.5** The voltage across an inductor leads the current through it by 90° .
- (a) True (b) False
- 9.6** The imaginary part of impedance is called:
- (a) resistance (b) admittance
 (c) susceptance (d) conductance
 (e) reactance
- 9.7** The impedance of a capacitor increases with increasing frequency.
- (a) True (b) False
- 9.8** At what frequency will the output voltage $v_o(t)$ in Fig. 9.39 be equal to the input voltage $v(t)$?
- (a) 0 rad/s (b) 1 rad/s (c) 4 rad/s
 (d) ∞ rad/s (e) none of the above

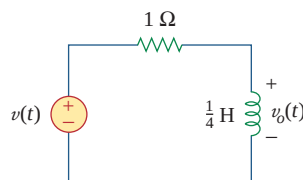


Figure 9.39
For Review Question 9.8.

- 9.9** A series RC circuit has $|V_R| = 12$ V and $|V_C| = 5$ V. The magnitude of the supply voltage is:
- (a) -7 V (b) 7 V (c) 13 V (d) 17 V
- 9.10** A series RCL circuit has $R = 30 \Omega$, $X_C = 50 \Omega$, and $X_L = 90 \Omega$. The impedance of the circuit is:
- (a) $30 + j140 \Omega$ (b) $30 + j40 \Omega$
 (c) $30 - j40 \Omega$ (d) $-30 - j40 \Omega$
 (e) $-30 + j40 \Omega$

Answers: 9.1d, 9.2c, 9.3b, 9.4b,d, 9.5a, 9.6e, 9.7b, 9.8d, 9.9c, 9.10b.

Problems

Section 9.2 Sinusoids

- 9.1** Given the sinusoidal voltage $v(t) = 50 \cos(30t + 10^\circ)$ V, find: (a) the amplitude V_m , (b) the period T , (c) the frequency f , and (d) $v(t)$ at $t = 10$ ms.
- 9.2** A current source in a linear circuit has
- $$i_s = 8 \cos(500\pi t - 25^\circ) \text{ A}$$
- (a) What is the amplitude of the current?
 (b) What is the angular frequency?
 (c) Find the frequency of the current.
 (d) Calculate i_s at $t = 2$ ms.
- 9.3** Express the following functions in cosine form:
- (a) $4 \sin(\omega t - 30^\circ)$ (b) $-2 \sin 6t$
 (c) $-10 \sin(\omega t + 20^\circ)$

9.4 Design a problem to help other students better understand sinusoids.



9.5 Given $v_1 = 20 \sin(\omega t + 60^\circ)$ and $v_2 = 60 \cos(\omega t - 10^\circ)$, determine the phase angle between the two sinusoids and which one lags the other.

9.6 For the following pairs of sinusoids, determine which one leads and by how much.

(a) $v(t) = 10 \cos(4t - 60^\circ)$ and $i(t) = 4 \sin(4t + 50^\circ)$

(b) $v_1(t) = 4 \cos(377t + 10^\circ)$ and $v_2(t) = -20 \cos 377t$

(c) $x(t) = 13 \cos 2t + 5 \sin 2t$ and $y(t) = 15 \cos(2t - 11.8^\circ)$

Section 9.3 Phasors

9.7 If $f(\phi) = \cos \phi + j \sin \phi$, show that $f(\phi) = e^{j\phi}$.

9.8 Calculate these complex numbers and express your results in rectangular form:

(a) $\frac{15 \angle 45^\circ}{3 - j4} + j2$

(b) $\frac{8 \angle -20^\circ}{(2 + j)(3 - j4)} + \frac{10}{-5 + j12}$

(c) $10 + (8 \angle 50^\circ)(5 - j12)$

9.9 Evaluate the following complex numbers and leave your results in polar form:

(a) $5 \angle 30^\circ \left(6 - j8 + \frac{3 \angle 60^\circ}{2 + j} \right)$

(b) $\frac{(10 \angle 60^\circ)(35 \angle -50^\circ)}{(2 + j6) - (5 + j)}$

9.10 Design a problem to help other students better understand phasors.



9.11 Find the phasors corresponding to the following signals:

(a) $v(t) = 21 \cos(4t - 15^\circ)$ V

(b) $i(t) = -8 \sin(10t + 70^\circ)$ mA

(c) $v(t) = 120 \sin(10t - 50^\circ)$ V

(d) $i(t) = -60 \cos(30t + 10^\circ)$ mA

9.12 Let $\mathbf{X} = 8 \angle 40^\circ$ and $\mathbf{Y} = 10 \angle -30^\circ$. Evaluate the following quantities and express your results in polar form:

(a) $(\mathbf{X} + \mathbf{Y})\mathbf{X}^*$

(b) $(\mathbf{X} - \mathbf{Y})^*$

(c) $(\mathbf{X} + \mathbf{Y})/\mathbf{X}$

9.13 Evaluate the following complex numbers:

(a) $\frac{2 + j3}{1 - j6} + \frac{7 - j8}{-5 + j11}$

(b) $\frac{(5 \angle 10^\circ)(10 \angle -40^\circ)}{(4 \angle -80^\circ)(-6 \angle 50^\circ)}$

(c) $\begin{vmatrix} 2 + j3 & -j2 \\ -j2 & 8 - j5 \end{vmatrix}$

9.14 Simplify the following expressions:

(a) $\frac{(5 - j6) - (2 + j8)}{(-3 + j4)(5 - j) + (4 - j6)}$

(b) $\frac{(240 \angle 75^\circ + 160 \angle -30^\circ)(60 - j80)}{(67 + j84)(20 \angle 32^\circ)}$

(c) $\left(\frac{10 + j20}{3 + j4} \right)^2 \sqrt{(10 + j5)(16 - j20)}$

9.15 Evaluate these determinants:

(a) $\begin{vmatrix} 10 + j6 & 2 - j3 \\ -5 & -1 + j \end{vmatrix}$

(b) $\begin{vmatrix} 20 \angle -30^\circ & -4 \angle -10^\circ \\ 16 \angle 0^\circ & 3 \angle 45^\circ \end{vmatrix}$

(c) $\begin{vmatrix} 1 - j & -j & 0 \\ j & 1 & -j \\ 1 & j & 1 + j \end{vmatrix}$

9.16 Transform the following sinusoids to phasors:

(a) $-10 \cos(4t + 75^\circ)$ (b) $5 \sin(20t - 10^\circ)$

(c) $4 \cos 2t + 3 \sin 2t$

9.17 Two voltages v_1 and v_2 appear in series so that their sum is $v = v_1 + v_2$. If $v_1 = 10 \cos(50t - \pi/3)$ V and $v_2 = 12 \cos(50t + 30^\circ)$ V, find v .

9.18 Obtain the sinusoids corresponding to each of the following phasors:

(a) $\mathbf{V}_1 = 60 \angle 15^\circ$ V, $\omega = 1$

(b) $\mathbf{V}_2 = 6 + j8$ V, $\omega = 40$

(c) $\mathbf{I}_1 = 2.8e^{-j\pi/3}$ A, $\omega = 377$

(d) $\mathbf{I}_2 = -0.5 - j1.2$ A, $\omega = 10^3$

9.19 Using phasors, find:

(a) $3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$

(b) $40 \sin 50t + 30 \cos(50t - 45^\circ)$

(c) $20 \sin 400t + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ)$

9.20 A linear network has a current input $4 \cos(\omega t + 20^\circ)$ A and a voltage output $10 \cos(\omega t + 110^\circ)$ V. Determine the associated impedance.

9.21 Simplify the following:

(a) $f(t) = 5 \cos(2t + 15^\circ) - 4 \sin(2t - 30^\circ)$

(b) $g(t) = 8 \sin t + 4 \cos(t + 50^\circ)$

(c) $h(t) = \int_0^t (10 \cos 40t + 50 \sin 40t) dt$

9.22 An alternating voltage is given by $v(t) = 20 \cos(5t - 30^\circ)$ V. Use phasors to find

$$10v(t) + 4 \frac{dv}{dt} - 2 \int_{-\infty}^t v(t) dt$$

Assume that the value of the integral is zero at $t = -\infty$.

9.23 Apply phasor analysis to evaluate the following.

(a) $v = 50 \cos(\omega t + 30^\circ) + 30 \cos(\omega t - 90^\circ)$ V

(b) $i = 15 \cos(\omega t + 45^\circ) - 10 \sin(\omega t + 45^\circ)$ A

9.24 Find $v(t)$ in the following integrodifferential equations using the phasor approach:

(a) $v(t) + \int v dt = 5 \cos(t + 45^\circ)$ V

(b) $\frac{dv}{dt} + 5v(t) + 4 \int v dt = 20 \sin(4t + 10^\circ)$ V

9.25 Using phasors, determine $i(t)$ in the following equations:

(a) $2 \frac{di}{dt} + 3i(t) = 4 \cos(2t - 45^\circ)$

(b) $10 \int i dt + \frac{di}{dt} + 6i(t) = 5 \cos(5t + 22^\circ)$ A

9.26 The loop equation for a series RLC circuit gives

$$\frac{di}{dt} + 2i + \int_{-\infty}^t i dt = \cos 2t \text{ A}$$

Assuming that the value of the integral at $t = -\infty$ is zero, find $i(t)$ using the phasor method.

9.27 A parallel RLC circuit has the node equation

$$\frac{dv}{dt} + 50v + 100 \int v dt = 110 \cos(377t - 10^\circ) \text{ V}$$

Determine $v(t)$ using the phasor method. You may assume that the value of the integral at $t = -\infty$ is zero.

Section 9.4 Phasor Relationships for Circuit Elements

9.28 Determine the current that flows through an 8- Ω resistor connected to a voltage source $v_s = 110 \cos 377t$ V.

9.29 What is the instantaneous voltage across a 2- μ F capacitor when the current through it is $i = 4 \sin(10^6 t + 25^\circ)$ A?

9.30 A voltage $v(t) = 100 \cos(60t + 20^\circ)$ V is applied to a parallel combination of a 40-k Ω resistor and a 50- μ F capacitor. Find the steady-state currents through the resistor and the capacitor.

9.31 A series RLC circuit has $R = 80 \Omega$, $L = 240$ mH, and $C = 5$ mF. If the input voltage is $v(t) = 10 \cos 2t$, find the current flowing through the circuit.

9.32 Using Fig. 9.40, design a problem to help other students better understand phasor relationships for circuit elements.

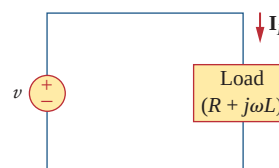


Figure 9.40

For Prob. 9.32.

9.33 A series RL circuit is connected to a 110-V ac source. If the voltage across the resistor is 85 V, find the voltage across the inductor.

9.34 What value of ω will cause the forced response v_o in Fig. 9.41 to be zero?

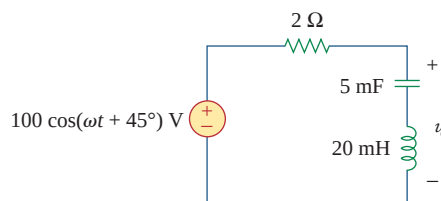


Figure 9.41

For Prob. 9.34.

Section 9.5 Impedance and Admittance

9.35 Find current i in the circuit of Fig. 9.42, when $v_s(t) = 50 \cos 200t$ V.

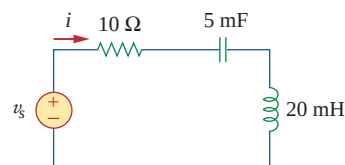


Figure 9.42

For Prob. 9.35.

- 9.36** Using Fig. 9.43, design a problem to help other students better understand impedance.

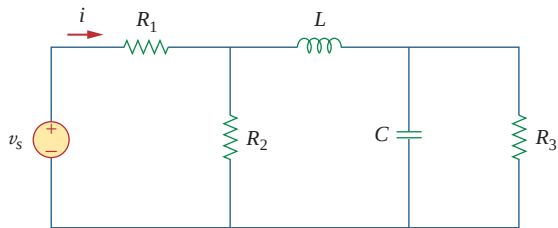


Figure 9.43

For Prob. 9.36.

- 9.37** Determine the admittance $\mathbf{Y}$ for the circuit in Fig. 9.44.

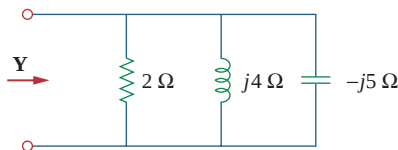


Figure 9.44

For Prob. 9.37.

- 9.38** Using Fig. 9.45, design a problem to help other students better understand admittance.

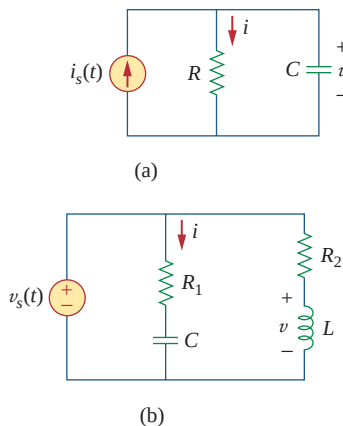


Figure 9.45

For Prob. 9.38.

- 9.39** For the circuit shown in Fig. 9.46, find Z_{eq} and use that to find current $\mathbf{I}$. Let $\omega = 10$ rad/s.

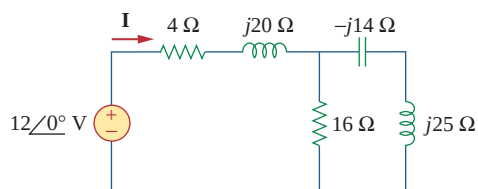


Figure 9.46

For Prob. 9.39.

- 9.40** In the circuit of Fig. 9.47, find i_o when:

- (a) $\omega = 1$ rad/s (b) $\omega = 5$ rad/s
(c) $\omega = 10$ rad/s

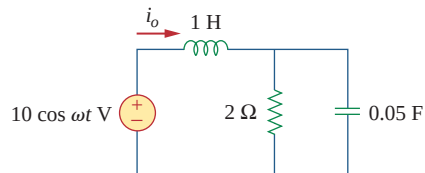


Figure 9.47

For Prob. 9.40.

- 9.41** Find $v(t)$ in the RLC circuit of Fig. 9.48.

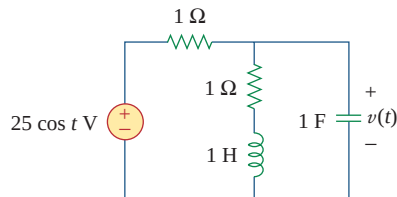


Figure 9.48

For Prob. 9.41.

- 9.42** Calculate $v_o(t)$ in the circuit of Fig. 9.49.

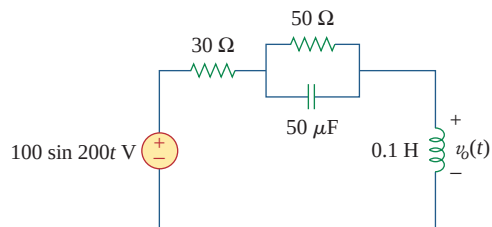


Figure 9.49

For Prob. 9.42.

- 9.43** Find current $\mathbf{I}_o$ in the circuit shown in Fig. 9.50.

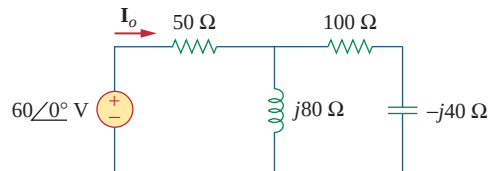


Figure 9.50

For Prob. 9.43.

- 9.44** Calculate $i(t)$ in the circuit of Fig. 9.51.

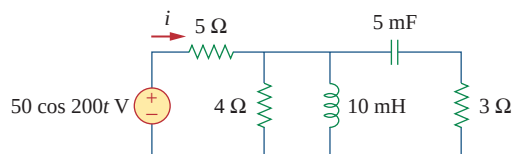


Figure 9.51

For prob. 9.44.

9.45 Find current I_o in the network of Fig. 9.52.

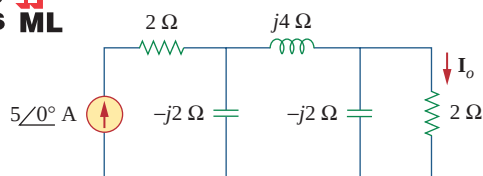


Figure 9.52

For Prob. 9.45.

9.46 If $i_s = 20 \cos(10t + 15^\circ)$ A in the circuit of Fig. 9.53, find i_o .

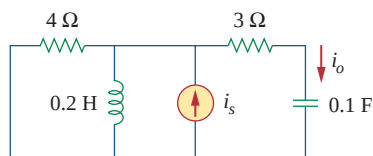


Figure 9.53

For Prob. 9.46.

9.47 In the circuit of Fig. 9.54, determine the value of $i_s(t)$.

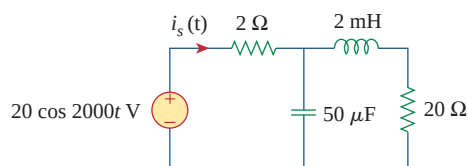


Figure 9.54

For Prob. 9.47.

9.48 Given that $v_s(t) = 20 \sin(100t - 40^\circ)$ in Fig. 9.55, determine $i_x(t)$.

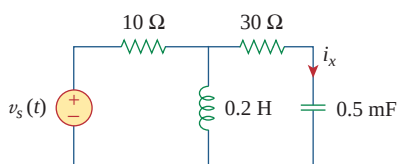


Figure 9.55

For Prob. 9.48.

9.49 Find $v_s(t)$ in the circuit of Fig. 9.56 if the current i_x through the 1-Ω resistor is $0.5 \sin 200t$ A.

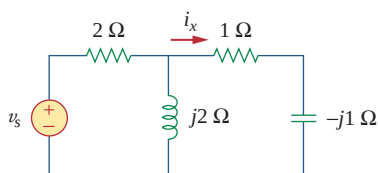


Figure 9.56

For Prob. 9.49.

9.50 Determine v_x in the circuit of Fig. 9.57. Let $i_s(t) = 5 \cos(100t + 40^\circ)$ A.

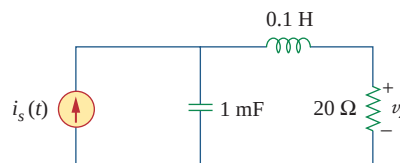


Figure 9.57

For Prob. 9.50.

9.51 If the voltage v_o across the 2-Ω resistor in the circuit of Fig. 9.58 is $-5 \cos 2t$ V, obtain i_s .

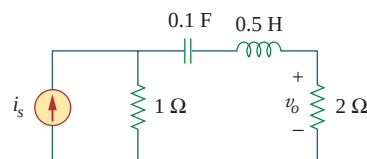


Figure 9.58

For Prob. 9.51.

9.52 If $V_o = 20\angle45^\circ$ V in the circuit of Fig. 9.59, find I_s .

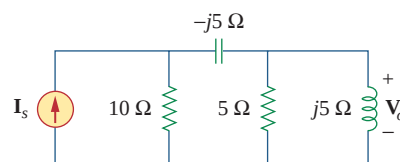


Figure 9.59

For Prob. 9.52.

9.53 Find I_o in the circuit of Fig. 9.60.

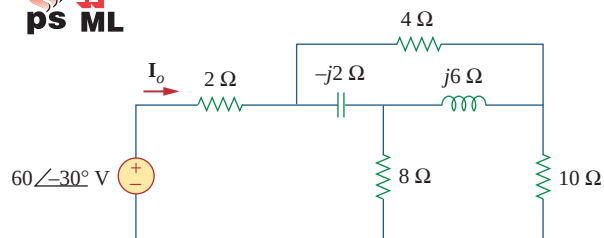


Figure 9.60

For Prob. 9.53.

9.54 In the circuit of Fig. 9.61, find V_s if $I_o = 2\angle0^\circ$ A.

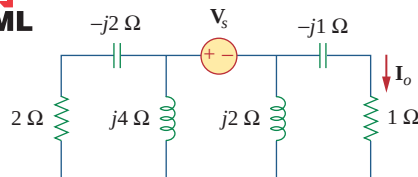


Figure 9.61

For Prob. 9.54.

- *9.55** Find $\mathbf{Z}$ in the network of Fig. 9.62, given that $\mathbf{V}_o = 8\angle 0^\circ \text{ V}$.

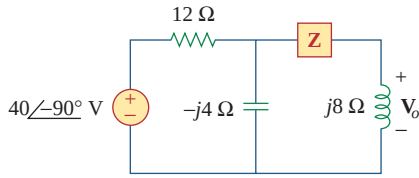


Figure 9.62

For Prob. 9.55.

Section 9.7 Impedance Combinations

- 9.56** At $\omega = 377 \text{ rad/s}$, find the input impedance of the circuit shown in Fig. 9.63.

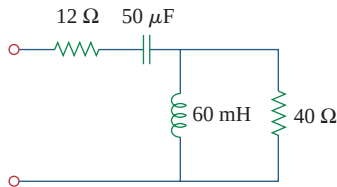


Figure 9.63

For Prob. 9.56.

- 9.57** At $\omega = 1 \text{ rad/s}$, obtain the input admittance in the circuit of Fig. 9.64.

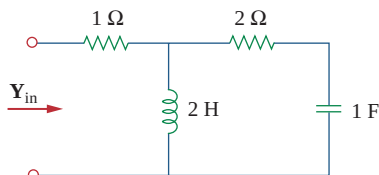


Figure 9.64

For Prob. 9.57.

- 9.58** Using Fig. 9.65, design a problem to help other students better understand impedance combinations.

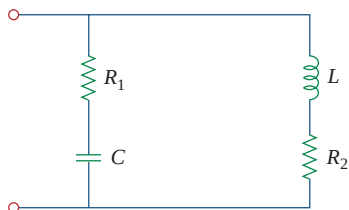


Figure 9.65

For Prob. 9.58.

* An asterisk indicates a challenging problem.

- 9.59** For the network in Fig. 9.66, find $\mathbf{Z}_{in}$. Let $\omega = 10 \text{ rad/s}$.

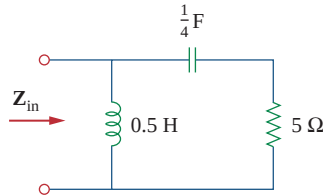


Figure 9.66

For Prob. 9.59.

- 9.60** Obtain $\mathbf{Z}_{in}$ for the circuit in Fig. 9.67.

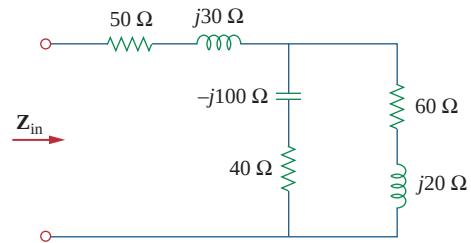


Figure 9.67

For Prob. 9.60.

- 9.61** Find $\mathbf{Z}_{eq}$ in the circuit of Fig. 9.68.

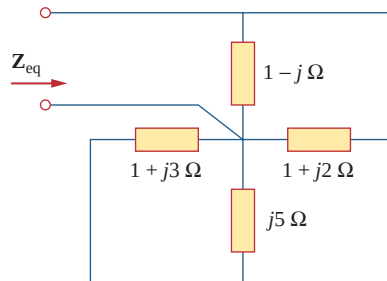


Figure 9.68

For Prob. 9.61.

- 9.62** For the circuit in Fig. 9.69, find the input impedance $\mathbf{Z}_{in}$ at 10 krad/s .

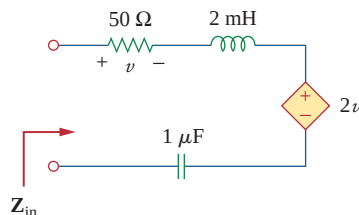


Figure 9.69

For Prob. 9.62.

9.63 For the circuit in Fig. 9.70, find the value of Z_T .

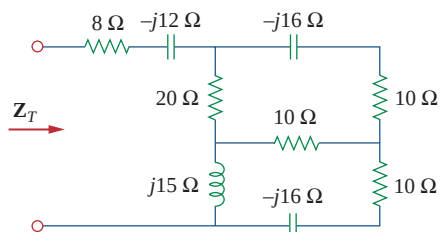


Figure 9.70

For Prob. 9.63.

9.64 Find Z_T and I in the circuit of Fig. 9.71.

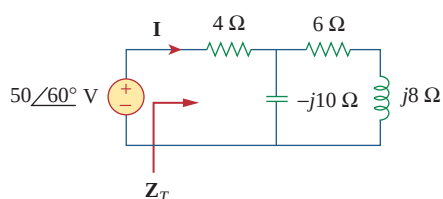


Figure 9.71

For Prob. 9.64.

9.65 Determine Z_T and I for the circuit in Fig. 9.72.

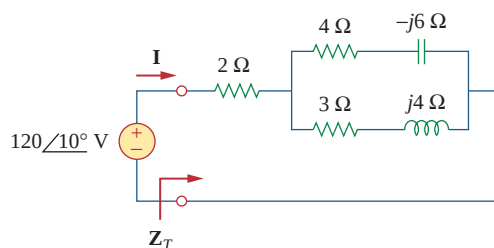


Figure 9.72

For Prob. 9.65.

9.66 For the circuit in Fig. 9.73, calculate Z_T and V_{ab} .

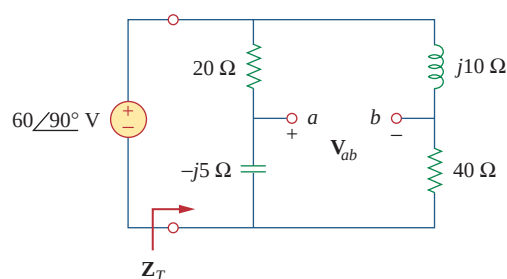
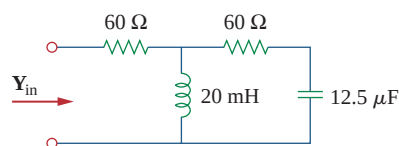


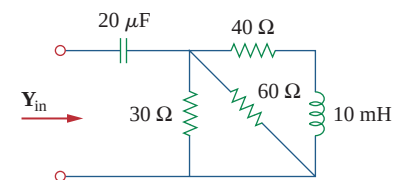
Figure 9.73

For Prob. 9.66.

9.67 At $\omega = 10^3$ rad/s, find the input admittance of each of the circuits in Fig. 9.74.



(a)



(b)

Figure 9.74

For Prob. 9.67.

9.68 Determine Y_{eq} for the circuit in Fig. 9.75.

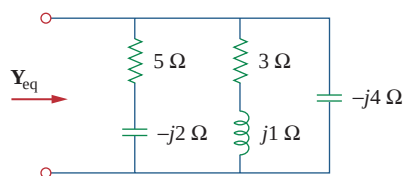


Figure 9.75

For Prob. 9.68.

9.69 Find the equivalent admittance Y_{eq} of the circuit in Fig. 9.76.

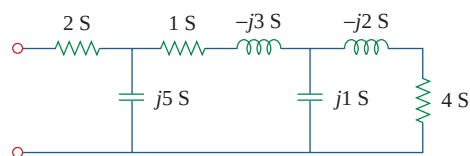


Figure 9.76

For Prob. 9.69.

9.70 Find the equivalent impedance of the circuit in Fig. 9.77.

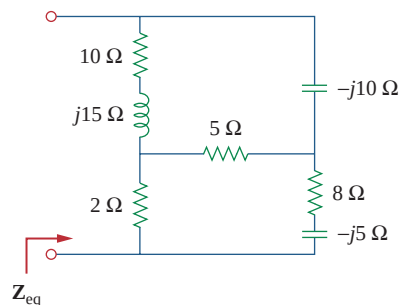


Figure 9.77

For Prob. 9.70.

- 9.71** Obtain the equivalent impedance of the circuit in Fig. 9.78.

ML

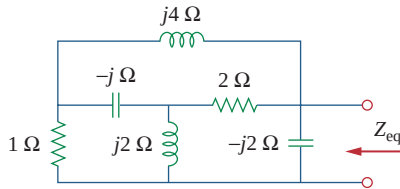


Figure 9.78

For Prob. 9.71.

- 9.72** Calculate the value of Z_{ab} in the network of Fig. 9.79.

ML

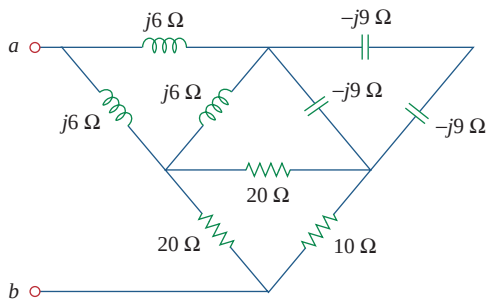


Figure 9.79

For Prob. 9.72.

- 9.73** Determine the equivalent impedance of the circuit in Fig. 9.80.

ML

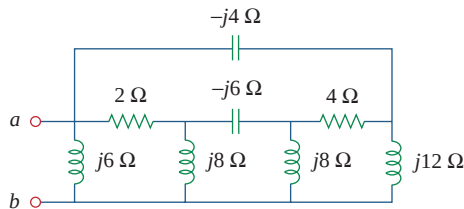


Figure 9.80

For Prob. 9.73.

Section 9.8 Applications

- 9.74** Design an RL circuit to provide a 90° leading phase shift.

ed

- 9.75** Design a circuit that will transform a sinusoidal voltage input to a cosinusoidal voltage output.

ed

- 9.76** For the following pairs of signals, determine if v_1 leads or lags v_2 and by how much.

- (a) $v_1 = 10 \cos(5t - 20^\circ)$, $v_2 = 8 \sin 5t$
 (b) $v_1 = 19 \cos(2t + 90^\circ)$, $v_2 = 6 \sin 2t$
 (c) $v_1 = -4 \cos 10t$, $v_2 = 15 \sin 10t$

- 9.77** Refer to the RC circuit in Fig. 9.81.

- (a) Calculate the phase shift at 2 MHz.
 (b) Find the frequency where the phase shift is 45° .

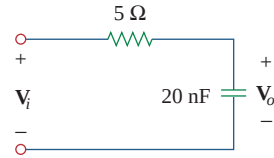


Figure 9.81

For Prob. 9.77.

- 9.78** A coil with impedance $8 + j6 \Omega$ is connected in series with a capacitive reactance X . The series combination is connected in parallel with a resistor R . Given that the equivalent impedance of the resulting circuit is $5 \angle 0^\circ \Omega$, find the value of R and X .

- 9.79** (a) Calculate the phase shift of the circuit in Fig. 9.82.
 (b) State whether the phase shift is leading or lagging (output with respect to input).
 (c) Determine the magnitude of the output when the input is 120 V.

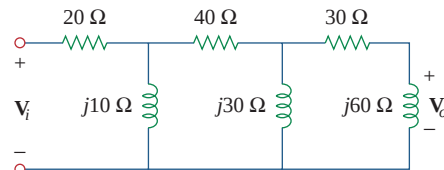


Figure 9.82

For Prob. 9.79.

- 9.80** Consider the phase-shifting circuit in Fig. 9.83. Let $V_i = 120$ V operating at 60 Hz. Find:

- (a) V_o when R is maximum
 (b) V_o when R is minimum
 (c) the value of R that will produce a phase shift of 45°

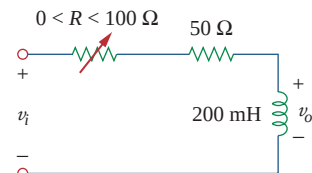


Figure 9.83

For Prob. 9.80.

- 9.81** The ac bridge in Fig. 9.37 is balanced when $R_1 = 400 \Omega$, $R_2 = 600 \Omega$, $R_3 = 1.2 \text{ k}\Omega$, and $C_2 = 0.3 \mu\text{F}$. Find R_x and C_x . Assume R_2 and C_2 are in series.

- 9.82** A capacitance bridge balances when $R_1 = 100 \Omega$, $R_2 = 2 \text{ k}\Omega$, and $C_s = 40 \mu\text{F}$. What is C_x , the capacitance of the capacitor under test?

- 9.83** An inductive bridge balances when $R_1 = 1.2 \text{ k}\Omega$, $R_2 = 500 \Omega$, and $L_s = 250 \text{ mH}$. What is the value of L_x , the inductance of the inductor under test?

- 9.84** The ac bridge shown in Fig. 9.84 is known as a *Maxwell bridge* and is used for accurate measurement of inductance and resistance of a coil in terms of a standard capacitance C_s . Show that when the bridge is balanced,

$$L_x = R_2 R_3 C_s \quad \text{and} \quad R_x = \frac{R_2}{R_1} R_3$$

Find L_x and R_x for $R_1 = 40 \text{ k}\Omega$, $R_2 = 1.6 \text{ k}\Omega$, $R_3 = 4 \text{ k}\Omega$, and $C_s = 0.45 \text{ }\mu\text{F}$.

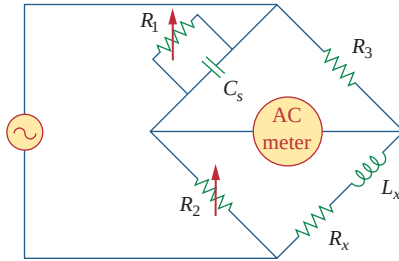


Figure 9.84
Maxwell bridge; For Prob. 9.84.

- 9.85** The ac bridge circuit of Fig. 9.85 is called a *Wien bridge*. It is used for measuring the frequency of a source. Show that when the bridge is balanced,

$$f = \frac{1}{2\pi \sqrt{R_2 R_4 C_2 C_4}}$$

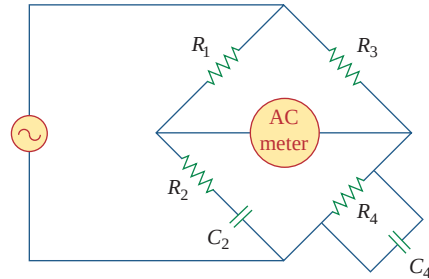


Figure 9.85
Wien bridge; For Prob. 9.85.

Comprehensive Problems

- 9.86** The circuit shown in Fig. 9.86 is used in a television receiver. What is the total impedance of this circuit?

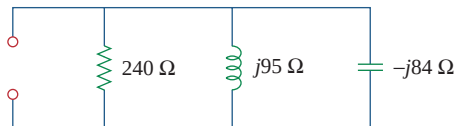


Figure 9.86
For Prob. 9.86.

- 9.87** The network in Fig. 9.87 is part of the schematic describing an industrial electronic sensing device. What is the total impedance of the circuit at 2 kHz?

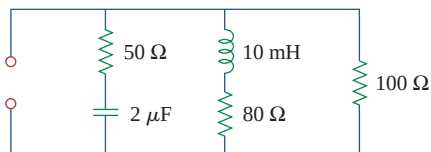


Figure 9.87
For Prob. 9.87.

- 9.88** A series audio circuit is shown in Fig. 9.88.
- What is the impedance of the circuit?
 - If the frequency were halved, what would be the impedance of the circuit?

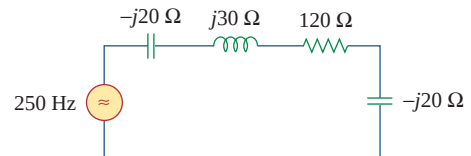


Figure 9.88
For Prob. 9.88.

- 9.89** An industrial load is modeled as a series combination of a capacitance and a resistance as shown in Fig. 9.89. Calculate the value of an inductance L across the series combination so that the net impedance is resistive at a frequency of 50 kHz.

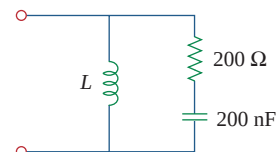


Figure 9.89
For Prob. 9.89.

- 9.90** An industrial coil is modeled as a series combination of an inductance L and resistance R , as shown in Fig. 9.90. Since an ac voltmeter measures only the magnitude of a sinusoid, the following

measurements are taken at 60 Hz when the circuit operates in the steady state:

$$|V_s| = 145 \text{ V}, \quad |V_1| = 50 \text{ V}, \quad |V_o| = 110 \text{ V}$$

Use these measurements to determine the values of L and R .

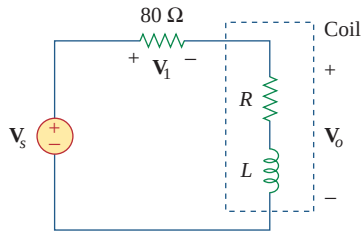


Figure 9.90

For Prob. 9.90.

- 9.91** Figure 9.91 shows a parallel combination of an inductance and a resistance. If it is desired to connect a capacitor in series with the parallel combination such that the net impedance is resistive at 10 MHz, what is the required value of C ?

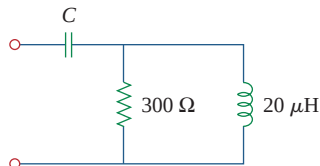


Figure 9.91

For Prob. 9.91.

- 9.92** A transmission line has a series impedance of $Z = 100 \angle 75^\circ \Omega$ and a shunt admittance of $Y = 450 \angle 48^\circ \mu\text{S}$. Find: (a) the characteristic impedance $Z_o = \sqrt{Z/Y}$, (b) the propagation constant $\gamma = \sqrt{ZY}$.

- 9.93** A power transmission system is modeled as shown in Fig. 9.92. Given the following;

Source voltage $V_s = 115 \angle 0^\circ \text{ V}$,
 Source impedance $Z_s = (2 + j) \Omega$,
 Line impedance $Z_\ell = (0.8 + j0.6) \Omega$,
 Load impedance $Z_L = (46.4 + j37.8) \Omega$,
 Find the load current I_L .

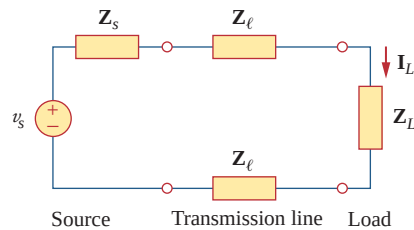


Figure 9.92

For Prob. 9.93.